

Reference teardown

Each reference was cloned and read at the code level, because one README promises features its code lacks and another repo holds three separate implementations. Every repo gave one or two patterns worth keeping and at least one mistake worth designing out.

Reference	Its job here	Take	Leave behind
3D-Portfolio	Frontend and AI architecture	React 19, React Three Fiber 9, drei and Tailwind 4 as the stack; a fixed canvas behind DOM text; a layered core object (wire shell, distorted core, particle halo) as vocabulary for an AI presence; pointer-driven easing	The Gemini key compiled into the browser bundle; a separate canvas per section; React state and timers inside the frame loop; new vectors every frame; a README promising a chat assistant the code lacks
openluma	Event and business logic	Visibility, approval and capacity resolving to approved, pending or waitlisted; token invites with expiry, email binding, rate-limited lookup and a transactional accept; a status timeline; host checks on every mutation; an AI agent whose risky tools need the user's approval and whose results render as cards; a cron with a constant-time secret; Event data with real IST offsets; a share image per event	Capacity checked and inserted in separate steps, which can oversell; instant waitlist promotion with no claim window or payment; an unsigned QR payload; chat history trusted from the browser; no payments

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EventNexus	Event-aware AI behaviour	Two concierge modes, global discovery and single-event answers; recommend only from the catalogue; quick-reply chips; full-screen chat on phones; a row lock around booking in its Spring Boot backend	The Gemini key in client code; the whole catalogue pasted into the prompt; a fallback that invents events and claims; a Next.js rewrite that books with no lock and no payment; waitlist seats handed out instantly, with no claim window
immersive-3d-web	Cinematic visual experience	The concept-to-form test; scroll progress applied inside the render loop; media queries for phones and reduced motion; instancing for particles; per-element micro-motion; locked states styled as an invite away; one-shot celebrations; pixel-ratio clamps; disposal; the 300 × 150 canvas fix	Generic polyhedra and slow idle spins as default motion
3js-avatar-chatbot	AI personality and presence	A structured turn, text plus a sentiment from a fixed list, driving expression; an event bus between chat and 3D; idle and talking states; audio level driving motion; typed errors mapped to status codes; an httpOnly session cookie	The human avatar; session memory in a server-side Map, which vanishes on serverless; a regex prompt-injection filter as the main defence; a turn that waits for full audio instead of streaming

Reference	Its job here	Take	Leave behind
react-three-fiber-examples	Production 3D engineering	React state for choices and refs for frames; a server-rendered page around a client canvas; adaptive pixel ratio and events; an on-demand frame loop for reduced motion; a WebGL check with a designed fallback; pausing when hidden; strict disposal; a QA checklist	Little else: the repo is mostly a guide and one example

What is new in Underdogs: none of the six has a human-reviewed invite queue, a timed location drop, a coin that doubles as invite and ticket, or a concierge built from the brand’s own mascot. Those come from Underdogs’ own launch material, not from the references.